In [3]:
```
import numpy as np
```

In [4]:
```
np.__version__
```

Out[4]:
```
'2.3.5'
```

In [5]:
```
my_list=[1,3,4,5,6,7,8,9,10]
```

In [6]:
```
my_list
```

Out[6]:
```
[1, 3, 4, 5, 6, 7, 8, 9, 10]
```

In [7]:
```
type(my_list)
```

Out[7]:
```
list
```

In [8]:
```
arr= np.array(my_list)
```

In [9]:
```
arr
```

Out[9]:
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [10]:
```
type(arr)
```

Out[10]:
```
numpy.ndarray
```

In [11]:
```
np.arange(12)
```

Out[11]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11])
```

In [12]:
```
np.arange(17,22)
```

Out[12]:
```
array([17, 18, 19, 20, 21])
```

In [13]:
```
np.arange(10, 20, 3)
```

Out[13]:
```
array([10, 13, 16, 19])
```

In [14]:
```
np.zeros(5)
```

Out[14]:
```
array([0., 0., 0., 0., 0.])
```

In [15]:
```
np.arange(10,30,5)  # 10- starting from 30- end point 5 - step count
```

Out[15]:
```
array([10, 15, 20, 25])
```

In [16]:
```
np.zeros(3) # parameter  tunning
```

Out[16]:
```
array([0., 0., 0.])
```

In [17]:
```
np.zeros(3, dtype=int) # hyperparameter  tunning
```

Out[17]:
```
array([0, 0, 0])
```

In [18]:
```
np.zeros((2,2), dtype=int )
```

Out[18]:
```
array([[0, 0],
       [0, 0]])
```

In [19]:
```
np.zeros((2,2), dtype=str)
```

Out[19]:
```
array([['', ''],
       ['', '']], dtype='<U1')
```

In [20]:
```
zero = np.zeros([2,2])
print(zero)

print('####')

print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
####
<class 'numpy.ndarray'>
```

In [21]:
```
np.zeros((5,10)) # bydefault 1arge -- will give row & 2nd arg - columns
```

Out[21]:
```
array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

In [22]:
```
np.ones(3)
```

Out[22]:
```
array([1., 1., 1.])
```

In [24]:
```
np.ones(3, dtype=int)
```

Out[24]:
```
array([1, 1, 1])
```

In [26]:
```
n=(6,7)
```

In [27]:
```
n
```

Out[27]:
```
(6, 7)
```

In [29]:
```
np.ones(n, dtype=int)
```

Out[29]:
```
array([[1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1, 1]])
```

In [31]:
```
np.twos(5) # only zeros and ones is work twos and threes and ll not allow
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[31], line 1
----> 1 np.twos(5) # only zeros and ones is work twos and threes and ll not allow

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

In [32]:
```
range(5)
```

Out[32]:
```
range(0, 5)
```

In [33]:
```
r= range(5)
```

In [34]:
```
r
```

Out[34]:
```
range(0, 5)
```

In [38]:
```
for i in r:
    print(i)
```

```
0
1
2
3
4
```

In [39]:
```
list (range(5))
```

Out[39]:
```
[0, 1, 2, 3, 4]
```

In [40]:
```
range(1,10)
```

Out[40]:
```
range(1, 10)
```

In [41]:
```
list(range(1,10))
```

Out[41]:
```
[1, 2, 3, 4, 5, 6, 7, 8, 9]
```

In [42]:
```
list(range(1,10,3))
```

Out[42]:
```
[1, 4, 7]
```

In [43]:
```
rand(3)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[43], line 1
----> 1 rand(3)

NameError: name 'rand' is not defined
```

In [44]:
```
random.rand(3)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[44], line 1
----> 1 random.rand(3)

NameError: name 'random' is not defined
```

In [46]:
```
np.random.rand(3)
```

Out[46]:
```
array([0.22618177, 0.6229257 , 0.96981565])
```

In [48]:
```
np.random.rand(3,4)
```

Out[48]:
```
array([[0.11042403, 0.12876355, 0.77199457, 0.72187575],
       [0.21280261, 0.610687  , 0.28816751, 0.65409489],
       [0.49829164, 0.42396983, 0.5005717 , 0.04540218]])
```

In [65]:
```
np.random.randint(2,5)
```

Out[65]:
```
4
```

In [69]:
```
np.random.randint(3)
```

Out[69]:
```
1
```

In [73]:
```
np.random.randint(-30,40,10)
```

Out[73]:
```
array([  8, -26,  -3,   4, -21, -15, -17, -14, -26,   8], dtype=int32)
```

In [82]:
```
np.random.randint(10,40,(10,10)) #generre the element 10 -30 with 4*4 mtri
```

Out[82]:
```
array([[18, 31, 25, 22, 16, 13, 34, 35, 39, 18],
       [22, 11, 35, 15, 30, 27, 13, 36, 30, 18],
       [28, 31, 16, 33, 26, 17, 27, 25, 18, 36],
       [22, 36, 32, 28, 34, 27, 39, 22, 19, 29],
       [19, 17, 19, 18, 38, 28, 26, 27, 23, 18],
       [21, 30, 25, 35, 24, 34, 18, 24, 39, 36],
       [36, 18, 17, 18, 35, 28, 32, 29, 24, 21],
       [38, 23, 37, 25, 23, 20, 10, 29, 12, 27],
       [37, 25, 27, 21, 12, 33, 38, 22, 28, 36],
       [27, 39, 34, 34, 28, 12, 25, 39, 22, 10]], dtype=int32)
```

In [86]:
```
a=np.arange(2,10).reshape(2,4)
```

In [87]:
```
a
```

Out[87]:
```
array([[2, 3, 4, 5],
       [6, 7, 8, 9]])
```

In [88]:
```
type(a)
```

Out[88]:
```
numpy.ndarray
```

In [105…
```
b=np.random.randint(1,40,(5,8))
```

In [106…
```
b
```

Out[106…
```
array([[ 5, 22, 24, 33,  3,  9, 34, 24],
       [16, 23, 32,  3, 15, 18, 39, 21],
       [30,  2, 30, 19, 39,  6, 15, 31],
       [29, 22, 37, 36, 14,  5, 24,  6],
       [20, 15, 36,  6, 18, 11, 31, 32]], dtype=int32)
```

In [107…
```
b[:]
```

Out[107…
```
array([[ 5, 22, 24, 33,  3,  9, 34, 24],
       [16, 23, 32,  3, 15, 18, 39, 21],
       [30,  2, 30, 19, 39,  6, 15, 31],
       [29, 22, 37, 36, 14,  5, 24,  6],
       [20, 15, 36,  6, 18, 11, 31, 32]], dtype=int32)
```

In [108…
```
b[0:3]
```

Out[108…
```
array([[ 5, 22, 24, 33,  3,  9, 34, 24],
       [16, 23, 32,  3, 15, 18, 39, 21],
       [30,  2, 30, 19, 39,  6, 15, 31]], dtype=int32)
```

In [111…
```
print(b[1,4])
```

```
15
```

In [114…
```
print(b[0,7])
```

```
24
```

In [115…
```
b
```

Out[115…
```
array([[ 5, 22, 24, 33,  3,  9, 34, 24],
       [16, 23, 32,  3, 15, 18, 39, 21],
       [30,  2, 30, 19, 39,  6, 15, 31],
       [29, 22, 37, 36, 14,  5, 24,  6],
       [20, 15, 36,  6, 18, 11, 31, 32]], dtype=int32)
```

In [116…
```
b[0]
```

Out[116…
```
array([ 5, 22, 24, 33,  3,  9, 34, 24], dtype=int32)
```

In [122…
```
b[4]
```

Out[122…
```
array([20, 15, 36,  6, 18, 11, 31, 32], dtype=int32)
```

In [123…
```
arr
```

Out[123…
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [124…
```
arr.max
```

Out[124…
```
<function ndarray.max>
```

In [125…
```
a=arr
```

In [126…
```
a
```

Out[126…
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [129…
```
print(a.max())
```

```
10
```

In [130…
```
print(a.min())
```

```
1
```

In [131…
```
print(a.mean())
```

```
5.888888888888889
```

In [132…
```
print(a.median())
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[132], line 1
----> 1 print(a.median())

AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

In [135…
```
from numpy import *
a
```

Out[135…
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [137…
```
print(median(a))  #Without work on import* can you please find the median, mode
```

```
6.0
```

In [138…
```
arr
```

Out[138…
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [146…
```
a=arr
a
```

Out[146…
```
array([ 1,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [148…
```
a.reshape(3,3)
```

Out[148…
```
array([[ 1,  3,  4],
       [ 5,  6,  7],
       [ 8,  9, 10]])
```

In [154…
```
a.reshape(3,3, order='F')  # print element with fortran
```

Out[154…
```
array([[ 1,  5,  8],
       [ 3,  6,  9],
       [ 4,  7, 10]])
```

In [153…
```
a.reshape(3,3, order='C') #A almost give you c type output
```

Out[153…
```
array([[ 1,  3,  4],
       [ 5,  6,  7],
       [ 8,  9, 10]])
```

In [152…
```
a.reshape(3,3, order='A')  #A almost give you c type output
```

Out[152…
```
array([[ 1,  3,  4],
       [ 5,  6,  7],
       [ 8,  9, 10]])
```

Indexing

In [163…
```
from numpy import *
mat = np.arange(0,100).reshape(10,10)
```

In [164…
```
mat
```

Out[164…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [165…
```
row=4
col=5
```

In [166…
```
row
```

Out[166…
```
4
```

In [167…
```
col
```

Out[167…
```
5
```

In [169…
```
print(mat[row,col])
```

```
45
```

In [170…
```
print(mat[4,5])
```

```
45
```

In [171…
```
mat
```

Out[171…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [172…
```
col =6
```

In [173…
```
col
```

Out[173…
```
6
```

In [174…
```
mat[7] # bybdefault it represent row
```

Out[174…
```
array([70, 71, 72, 73, 74, 75, 76, 77, 78, 79])
```

In [175…
```
mat [:,7]    # With Slices
```

Out[175…
```
array([ 7, 17, 27, 37, 47, 57, 67, 77, 87, 97])
```

In [176…
```
mat[row,:]   #row=4
```

Out[176…
```
array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

In [177…
```
mat
```

Out[177…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [182…
```
mat[5:7]
```

Out[182…
```
array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

In [180…
```
mat[0:10:3]
```

Out[180…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [183…
```
mat[::-3]
```

Out[183…
```
array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

In [184…
```
mat
```

Out[184…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [186…
```
mat[2:3,2:4]
```

Out[186…
```
array([[22, 23]])
```

In [189…
```
print(mat[1,2])
```

```
12
```

In [199…
```
mat[::-1]
```

Out[199…
```
array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

In [200…
```
mat[2:4,3:4]
```

Out[200…
```
array([[23],
       [33]])
```

In [202…
```
mat[3:5,2:4]
```

Out[202…
```
array([[32, 33],
       [42, 43]])
```

In [203…
```
mat
```

Out[203…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

Masking / Filter

In [204…
```
mat
```

Out[204…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [205…
```
a=mat
a
```

Out[205…
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [206…
```
a2 = a<50
```

In [207…
```
a2
```

Out[207…
```
array([[ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False]])
```

In [212…
```
a3 =a[a<50]
```

In [213…
```
a3
```

Out[213…
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

In [215…
```
a4 = a3[a3==45]
```

In [216…
```
a4
```

Out[216…
```
array([45])
```

In [219…
```
a5= a[a>=23]
```

In [220…
```
a5
```

Out[220…

```
array([23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39,
       40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56,
       57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73,
       74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90,
       91, 92, 93, 94, 95, 96, 97, 98, 99])
```

In [ ]: